

Quantum Computing Paradigms in Modern Power Systems

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Abstract

This review report provides an analysis of academic publications regarding Quantum Algorithms in the context of Power Systems problems. These works investigate the integration of quantum machine learning (QML), gate-based quantum computing, and adiabatic quantum annealing (QA) into power system operations. Specifically, they address continuous power flow analysis, state estimation, electromagnetic transient simulations, and mixed-integer combinatorial optimizations such as Stochastic Unit Commitment (SUC). I have tried to compare my personal summaries with the original texts, highlight and correct critical conceptual misunderstandings, and present an appendix outlining the foundational concepts that bridge classical power system operations and quantum hardware implementations. I recommend referring to the appendix while reading this main report to better understand the underlying concepts.

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1 Introduction

Modern electrical power grids are experiencing unprecedented operational challenges driven by the aggressive integration of volatile distributed energy resources (DERs) and low-inertia inverter-based resources (IBRs). Traditional classical algorithms, which scale polynomially with system size, are hitting hard computational limits when performing real-time static and transient analyses under high scenario uncertainty.

Quantum computing has emerged as a revolutionary computational paradigm capable of bypassing these bottlenecks. This report synthesizes the literature concerning both gate-based and adiabatic quantum computing applied to power systems, ordering the papers chronologically by publication date. For context, while conceptually slightly different, adiabatic computing and quantum annealing are treated synonymously in this report. To a control engineer, quantum annealing can be intuitively understood as allowing a physical dynamical system to naturally relax into its minimum-energy equilibrium state, which encodes the optimal solution to a given problem.

2 Review of the Literature

2.1 Zhou et al. (2022) – Quantum Computing in Power Systems

Original Publication Date: Received June 3, 2022; Accepted June 21, 2022.

2.1.1 Detailed Summary

Zhou et al. present a comprehensive survey introducing the framework of Quantum-Engineered Smart Grids (QGrids), mapping key power system operation perspectives to quantum algorithms. The paper categorizes the application of quantum algorithms into static analysis, transient analysis, stochastic analysis, optimization, stability, and control.

- **Static Analysis (Power Flow and State Estimation):** Traditional AC Power Flow (PF) analysis relies on iterative classical solvers like the Newton-Raphson (NR) method. The core computational bottleneck of the NR method is the inversion of the Jacobian matrix (J), which scales classically as $\mathcal{O}(N^3)$ where N is the number of buses—a scaling issue familiar in the computation of large state-transition matrices in control systems. The authors propose a hybrid pipeline using the Variational Quantum Linear Solver (VQLS) to solve the system matrix equation $J\Delta x = \Delta y$ inside the NR loop, reducing the gate-depth overhead compared to traditional fault-tolerant algorithms.

For State Estimation (SE), the authors discuss filtering noisy SCADA and PMU sensor data. While SCADA measurements are slow and asynchronous, PMUs provide ultra-fast, GPS-synchronized phasor measurements of phase angles (δ). The paper reviews how Quantum Neural Networks (QNNs) can directly ingest PMU time-series vectors to perform sub-millisecond classification of transient faults.

- **Transient Analysis:** Electromagnetic Transients Program (EMTP) waveform tracing requires resolving discretized differential-algebraic equations at microsecond time steps. The paper reviews Quantum EMTP (QEMTP) algorithms based on the HHL algorithm and VQLS. To bypass the tiny time steps necessitated by high-frequency 60 Hz carriers, Shifted Frequency Analysis (SFA) is integrated with quantum algorithms (QSFA). This shifts the signal down to a 0 Hz baseband, allowing for larger time-steps and solving the resulting envelope matrix equations via VQLS.

- **Optimization and Distributed Control:** Combinatorial problems such as Unit Commitment (UC) are mathematically formulated as Quadratic Unconstrained Binary Optimization (QUBO) problems, making them natively suitable for Quantum Annealers or the gate-based Quantum Approximate Optimization Algorithm (QAOA). Additionally, the paper reviews Quantum ADMM (Q-ADMM) pipelines for coordinate optimization, and Quantum Generative Adversarial Networks (QGANs) to generate realistic synthetic scenarios modeling renewable volatility.

2.2 Pelofske et al. (2023) – Quantum Annealing vs. QAOA: 127 Qubit Higher-Order Ising Problems on NISQ Computers

Original Publication Date: 2023 (Presented at the International Conference on High Performance Computing).

2.2.1 Detailed Summary

Pelofske et al. perform a direct hardware-to-hardware comparison between adiabatic quantum annealing (QA) and the gate-based Quantum Alternating Operator Ansatz/Quantum Approximate Optimization Algorithm (QAOA). The authors construct randomly generated minimization Ising models that natively match both the heavy-hexagonal lattice of IBM’s 127-qubit *ibm_washington* transmon processor and the Pegasus topology of D-Wave Advantage chips (Advantage system 4.1 and 6.1). Crucially, the problems are designed to incorporate cubic *ZZZ* (higher-order) interactions. Key findings from this large-scale empirical evaluation include:

1. **QA Dominance:** Quantum Annealing natively outperforms all tested variations of QAOA (1 and 2 rounds, with and without error mitigation) across all 10 problem instances, consistently settling into lower energy states closer to the global optimum.
2. **QAOA Validation:** Despite being outperformed by QA, both 1-round and 2-round QAOA runs yield significantly better objective values than 10,000 classical random samples. This proves that short-depth variational circuits can maintain quantum coherence on noisy processors before decohering into meaningless static.
3. **Dynamical Decoupling Efficacy:** Incorporating digital Dynamical Decoupling (DDD)—specifically applying rapid *X-X* microwave pulse sequences to idle qubits scheduled via the “As Late As Possible” (ALAP) algorithm—successfully cancels background environmental noise. From a control perspective, this acts as a highly effective open-loop control sequence designed for disturbance rejection. DDD enables the 2-round QAOA to marginally outperform the 1-round QAOA, which would otherwise be impossible due to accumulative gate errors.

2.3 Tripathi et al. (2024) – A Comprehensive Survey on Quantum Annealing

Original Publication Date: December 3, 2024.

2.3.1 Detailed Summary

Tripathi et al. provide a comprehensive survey outlining the milestones, hardware architectures, and industry adoption of Quantum Annealing (QA) up to late 2024. The paper establishes a benchmarking framework comparing QA with classical algorithms across metrics such as Time-to-Target (TTT), success probability, energy consumption, and hardware scalability.

- **Chronological Journey:** The survey maps out critical milestones, including the theoretical introduction of QA in 1989, the founding of D-Wave Systems in 2000, early physical

prototypes, the quantum speedup debates of 2013-2015, and scaling up to the 5000-qubit Advantage system in 2020.

- **Industrial Use Cases:** The authors examine active commercial deployments utilizing Quantum Computing as a Service (QCaaS), such as traffic routing (Volkswagen), telecom optimization (NTT DOCOMO), and supply chain scheduling (BASF/SAP). The survey emphasizes that 100% of these enterprise solutions are hybrid quantum-classical. They rely on supercomputers for the bulk of data processing and use QA QPUs strictly as co-processors to accelerate NP-hard combinatorial bottlenecks.
- **Strategic Hardware Trade-offs:** While QA displays exponential speedups in TTT for rugged energy landscapes and operates with lower electricity consumption than classical clusters, it remains severely restricted by:
 1. *Partial Qubit Connectivity:* Current topologies require complex "minor embedding" techniques. This involves grouping multiple physical qubits into a single logical variable chain, which wastes physical qubit capacity and reduces computational precision.
 2. *Analog Noise and Limited Coherence:* Due to the lack of active error correction, environmental thermal fluctuations can easily corrupt the fragile quantum annealing state, degrading success probabilities as problem sizes scale.
- **Roadmap Solution:** The survey explicitly concludes that the most viable near-term engineering path relies on hybrid decomposition methods (such as Benders' Decomposition and ADMM) to mathematically partition large problems into quantum-manageable subproblems.

2.4 Hong et al. (2026) – Qubit-Efficient Quantum Annealing for Stochastic Unit Commitment

Original Publication Date: August 11, 2026 (IEEE Transactions on Power Systems).

2.4.1 Detailed Summary

Hong et al. tackle the Stochastic Unit Commitment (SUC) problem, classically formulated as an NP-hard Mixed-Integer Programming (MIP) problem. SUC optimizes the day-ahead hourly ON/OFF status ($u_{g,t}$) of N thermal generators across T hours while minimizing operational costs and accommodating renewable generation fluctuations across dozens of weather scenarios (h).

When solved via a classical Benders Decomposition (BD) framework, the first-stage master problem remains highly demanding for classical CPUs, whereas the independent second-stage scenario subproblems can be solved efficiently in parallel. This paper investigates offloading the master problem to a Quantum Annealer. However, the authors demonstrate that a standard quantum formulation of the Benders master problem falls into the "Qubit Trap":

$$\text{Total Qubits (Standard QA)} = NT + J + MP + kF \quad (1)$$

where NT represents the useful binary decisions, J the binary-encoded continuous lower bound variables, MP the slack variables required to handle structural inequality constraints, and kF the auxiliary slack variables that grow linearly with each Benders iteration k . On limited hardware, this linear qubit scaling causes immediate system crashes. To resolve this, the authors propose the hybrid **QPHR-ADMM** algorithm, presenting three key mathematical tools:

1. **Powell-Hestenes-Rockafellar Augmented Lagrangian Multiplier (PHR-ALM) Integration:** QPHR-ALM incorporates inequality constraints directly into the objective function. By setting the first derivative of the Lagrangian with respect to continuous slack variables to zero, the authors derive an analytical identity for the slack variables based solely on Lagrange multipliers (λ) and penalty weights (σ). This completely deletes the physical slack variables, eliminating the MP and kF terms, keeping the qubit overhead at a constant $NT + J$, and breaking the iteration dependency.
2. **Unit-wise Direct-extended ADMM (D-ADMM) Decomposition:** To prevent qubit requirements from scaling with the number of units (N), the master problem is decomposed unit-wise using D-ADMM. Each single generator’s status over the 24-hour horizon represents a block of 24 variables, which is solved sequentially (serially) on the QPU. This limits the physical qubit requirements of the QPU to a constant $\max\{T, J\}$ (e.g., 24 qubits).
3. **Empirical Demonstration:** The algorithm was executed on the D-Wave Advantage system 4.1 using real-world testing systems up to the IEEE 118-bus system (comprising 1,296 binary variables). The QPHR-ADMM successfully solved the master problem using only 24 qubits, converging to solutions within 0.31% of classical Gurobi MIP solvers, establishing an ultra-scalable framework for resource-constrained quantum hardware.

An examination of the timing breakdown reveals a massive latency discrepancy:

- Direct QPU access time (T_{QPU}): 22.94 seconds (where pure annealing time, T_{anneal} , is only 1.61 seconds).
- Classical Orchestration latency (T_{orch}): 370.53 seconds.

The total wall-clock time is entirely dominated by classical API overhead (communication, minor-embedding, and queue times) rather than quantum processing.

2.5 Kaseb et al. (2026) – Performance Comparison of Gate-Based and Adiabatic Quantum Computing for AC Power Flow Problem

Original Publication Date: 2026

2.5.1 Detailed Summary

Kaseb et al. present the first direct comparative study between gate-based quantum computing (using PennyLane’s *lightning.qubit* simulator to execute QAOA) and adiabatic quantum computing (using D-Wave’s physical Quantum Annealer and Fujitsu’s specialized Digital Annealer, QIO) for solving the continuous, non-linear, and non-convex AC Power Flow (PF) equations. The paper outlines the following workflow:

- **Combinatorial Power Flow Formulation:** The continuous AC PF equations are rewritten as a combinatorial optimization problem by minimizing the sum of squared active and reactive power mismatches:

$$\min_{\mathbf{s}} \sum_{i=1}^N (P_i - P_{i,\text{target}})^2 + (Q_i - Q_{i,\text{target}})^2 \quad (2)$$

- **Iterative Base-Value Refinement (The Kaseb Trick):** Because Ising machines only handle binary spins ($s_i \in \{\pm 1\}$), the continuous variables (the real part μ and imaginary part ω of the voltage) must be discretized—a process conceptually similar to digital signal

quantization in control systems. Instead of mapping them to long, resource-heavy binary strings, the algorithm uses only 1 spin decision variable per continuous value. The spin dictates whether to add or subtract a step size (Δ) from the previous base value (μ_0, ω_0):

$$\mu_i = \mu_0^i + s_i^\mu \Delta \mu_i, \quad \omega_i = \omega_0^i + s_i^\omega \Delta \omega_i \quad (3)$$

After each iteration, the classical CPU updates the base values ($\mu_0 \leftarrow \mu$) and exponentially shrinks the step size ($\Delta \rightarrow 0.1 \times \Delta$). This effectively zooms into the solution space, shifting from coarse global searching to fine 10^{-3} convergence.

• **Comparative Performance Benchmarks (4-Bus Test System):**

1. *Fujitsu QIIO (Specialized CMOS DA)*: Achieved the fastest convergence, requiring only 63 iterations (average iteration time: 0.06 seconds, residual error: 3.31×10^{-4}).
2. *D-Wave QA (Physical Annealer)*: Required 222 iterations to satisfy the 10^{-3} convergence threshold. It showed ultra-fast QPU access time (0.015 seconds per iteration) but suffered from a wall-clock iteration time of 1.25 seconds due to minor-embedding and communications overhead.
3. *Gate-Based QAOA (Simulator)*: Representing the 6 unknown variables required 8 qubits on a statevector simulator. QAOA failed to converge to the predefined threshold within 300 iterations, primarily bottlenecked by classical parameter optimization (100 Adam gradient steps per iteration).

3 Summary Matrix of the Reviewed Literature

Table 1 summarizes the key characteristics, methodologies, and performance metrics of the reviewed papers in chronological order.

Table 1: Summary Matrix of Quantum Power Systems Literature

Study	Computing Paradigm	Target Grid Problem	Core Methodology	Key Findings / Performance
Zhou et al. (2022)	Gate-Based & QA	PF, SE, EMTP, UC	Hybrid VQLS, QSFA, QNN, Q-ADMM	Pioneer survey; established QGrids.
Pelofske et al. (2023)	Gate-Based vs. QA	Heavy-Hex Ising Models	QAOA (X -mixer) vs. QA on Pegasus	QA outperformed QAOA; DDD improves QAOA.
Tripathi et al. (2024)	Adiabatic QA	Logistics, Portfolio, Scheduling	Benchmark framework, TTT, minor embedding	Survey; hybrid algorithms are the path forward.
Hong et al. (2026)	Adiabatic QA (D-Wave)	Stochastic Unit Commitment	BD, QPHR-ALM, D-ADMM	Constant 24-qubit footprint; 0.31% cost gap.
Kaseb et al. (2026)	Gate vs. Adiabatic	AC Power Flow Problem	Iterative Base-Value Refinement	QHIO was fastest; QAOA struggled to converge.